

Learned representations

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Debugging shapes and gradients

Without opening the old lesson, explain one key idea from Debugging shapes and gradients. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Regularisation and early stopping

Without opening the old lesson, explain one key idea from Regularisation and early stopping. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Batching and epochs

Without opening the old lesson, explain one key idea from Batching and epochs. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Prove an information loss for Learned representations. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Learned representations. Proposed explanation: “Two hidden values preserve both input measurements”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Audit a new representation independently

A different layer uses $g = [\text{ReLU}(\text{width} - \text{height} - 1), \text{ReLU}(\text{height} - \text{width} - 1)]$. Encode $[4, 3]$ and $[3, 4]$. State their targets for width strictly greater than height. Can any deterministic head receiving only g classify both correctly?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Trace new inputs

Encode $[8, 3]$, $[3, 8]$ and $[4, 4]$ with the lesson layer. Apply its prediction rule and explain the square boundary.

My reasoning and evidence

6. W2 Prove an information loss

Give the hidden vectors and areas for $[4, 2]$ and $[5, 3]$. Explain why increasing only the complexity of a deterministic area head cannot make both exact predictions correct.

My reasoning and evidence

7. W3 Compare another encoding

The alternative representation is $s = \text{width} + \text{height}$. Use $[4, 2]$ and $[2, 4]$ to decide whether s alone always supports the wider-than-height task.

My reasoning and evidence

8. W4 Audit a new representation independently

A different layer uses $g = [\text{ReLU}(\text{width} - \text{height} - 1), \text{ReLU}(\text{height} - \text{width} - 1)]$. Encode $[4, 3]$ and $[3, 4]$. State their targets for width strictly greater than height. Can any deterministic head receiving only g classify both correctly?

My reasoning and evidence

9. W5 Repair and qualify a claim

For $h = [2, 0]$ and $s = 8$, recover width, height and area. Does this algebra prove that a training program learned the representation or that it will work well on noisy measurements?

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled learned representations evidence sheet to Image-pattern Network Investigation. Show how you can trace an input through a hidden representation and a decision rule. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Learned representations

See when a representation preserves a distinction and when it destroys it.

Toy input [left, right]; representation $h = [\text{ReLU}(\text{left} - \text{right}), \text{ReLU}(\text{right} - \text{left})]$. $A = [5, 2]$; $B = [2, 5]$; target is left-brighter versus right-brighter.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

A representation can preserve direction

	left	right	h1	h2
A	5	2	3	0
B	2	5	0	3

Read a cell using both its row and column labels.

Learned representations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Encode A

A becomes $[3, 0]$, with left brighter.



2. Encode B

B becomes $[0, 3]$, with right brighter.



3. Compare collapsed summary

If the head receives only $h_1 + h_2$, both become 3 and the target distinction is lost.

CHANGE ONE THING

The head keeps both ordered hidden values instead of their sum.

Explain the changed result using the matching lesson.

Why can no deterministic head using only the sum classify both correctly? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

